Vasudev Malyan

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Education

Jul 2019 – Aug 2026

MSc - PhD in Environmental Science and Engineering

Indian Institute of Technology Bombay, Mumbai, India, *MSc Grade: 9.68/10*

- Received the **Prime Minister's Research Fellowship** (2023–25), awarded to a small national cohort of doctoral students across Indian institutions
- Received the **Late Shri Girish Vishnuprasad Desai Award** (2023) for Academic Excellence

Jul 2016 – May 2019

BSc in Physical Science

Kirori Mal College, University of Delhi, Delhi, India, *BSc Grade: 8.21/10*

Professional Experience

Jul 2026 – Present

Technical Expert — Air Quality Sensor Technologies, Ubiqedge Technologies Pvt. Ltd., India

- Heading **R&D**, defining the technical roadmap and research priorities for **sensor calibration, source apportionment**, and **new product development** within the air quality sensing portfolio
- Leading development of **network-scale calibration models** and **source characterization** for an ongoing low-cost sensor deployment across **150+ construction sites** in Maharashtra, India
- Developing a **source-informed calibration engine** that classifies site typology from sensor-native features and applies source-stratified calibration

Jan 2026 – May 2026

Research Associate, Indian Institute of Technology Bombay, Mumbai, India

- Built a **physics-based calibration framework** for low-cost sensors that reduces dependence on co-located reference monitors, tested from chamber to field deployment
- Developed a **probabilistic Mie-scattering model** coupling κ-Köhler hygroscopic growth with sensor-specific optical response, enabling calibration without co-located reference monitors
- Validated the framework against **chamber experiments** and **field deployments** spanning multiple aerosol types and humidity regimes

Research Experience

Jul 2021 – May 2026

PhD Project: Calibration of low-cost particulate matter sensors using physics-based and machine learning frameworks — from laboratory evaluation to field applications

PI: Prof. Manoranjan Sahu, Indian Institute of Technology Bombay, India

- Completed PhD with **5 first-author** and **7 co-authored publications**
- Performed **controlled chamber experiments** across aerosol types, size distributions, and humidity regimes to isolate the physical mechanisms driving sensor error, published as a two-part series in *ACS ES&T Air* assessing both number- and mass-concentration response
- Developed a **physical-optical model** coupling Mie scattering with κ-Köhler hygroscopic growth theory, an inexpensive route to calibrating sensors without a co-located regulatory monitor
- Established that **aerosol source** and **size distribution** govern whether a calibration model transfers between sites using data from **field campaigns** across NCT-Delhi and Mumbai
- Experienced with **aerosol instrumentation** (SMPS-CPC, BAM-1020, GRIMM 11-D and OPS 1.109, TSI DustTrak, PurpleAir PA-II and Alphasense OPC-N2/N3), and **generation** (PALAS Aerosol Dust Generator, TSI Jet Nebulizer) systems
- Extended the work into policy, co-authoring a **T20 brief** on sensor-satellite synergies for addressing the issues related to air quality monitoring in G20 countries
- **Oral presenter** at the National Environment Conference (2024), Indian Aerosol Science and Technology Association Conference (2023), and Air Sensors International Conference (2022)
- Invited **peer reviewer** for **Atmospheric Pollution Research**

Feb 2022 – Apr 2023

IoT-Based Low-cost Sensor Network for Air Quality Monitoring

PI: Prof. Manoranjan Sahu, Indian Institute of Technology Bombay, India

- Designed and implemented the **IoT-based LCS network architecture** across the IIT Bombay campus, developed **calibration algorithms**, and integrated them into the project software
- Managed collaboration between the software development team and institute stakeholders

Dec 2021 – Apr 2022

Pilot Study on Reducing Particulate Air Pollution in Urban Areas Using an Air Cleaning System ("Smog Tower")

PI: Prof. Manoranjan Sahu, Indian Institute of Technology Bombay, India

- Built an **end-to-end analysis pipeline** integrating reference monitors and low-cost sensors to evaluate impact of smog-tower on hyperlocal air quality at two sites in NCT-Delhi
- Designed automated worksheets to process and visualize intrinsic and extrinsic filter differential pressure and track performance targets of the smog towers

Jul 2019 – Jul 2021

MSc Project: Techno-economic study of air pollution control technologies for capturing particulate matter

PI: Prof. Manoranjan Sahu, Indian Institute of Technology Bombay, India

- Evaluated **industrial particulate control technologies** including electrostatic precipitators, baghouse filters, cyclones, and scrubbers against capture efficiency, capital cost, and operating cost to produce selection guidance for industrial abatement
- Built the analysis around **particle size-resolved collection efficiency** rather than bulk removal rates, so that technology choice could be matched to the emitting process
- Extended the review to **indoor PM control**, comparing filtration and air-cleaning technologies for occupational settings, published in *Aerosol Science and Engineering*
- Master's seminar report published as a **book chapter with Springer Nature** on technologies for controlling particulate emissions from industry
- Achieved **highest CPI** in the MSc batch (9.68/10), recognized with the Late Shri Girish Vishnuprasad Desai Award for Academic Excellence

Publications

13 total, 6 first-author, citations = 154, h-index = 7, i10-index = 5

Peer-reviewed

Malyan, V., & Sahu, M., "Development and Assessment of a Physical-optical Calibration Model for Low-cost Sensors", *Manuscript under review* (2026).

Malyan, V., Sahu, M., & Nawale, P., "Design and Evaluation of a Calibration Chamber for Low-Cost PM Sensors – Part 2: Mass Concentration-Based Assessment", *ACS ES&T Air*, 3(6), 1677–1690 (2026). (Link)

Malyan, V., Nawale, P., & Sahu, M., "Design and Evaluation of a Calibration Chamber for Low-Cost PM Sensors – Part 1: Number Concentration-Based Assessment", *ACS ES&T Air*, 3(1), 129–141 (2025). (Link)

Zaid, M., Nawale, P., Kumar, V., **Malyan, V.**, & Sahu, M., "Investigating IoT-Based Low-cost Sensor Network for Real-Time Hyper-Local Air Quality Monitoring and Exposure Assessment", *Atmospheric Pollution Research*, 17(2), 102749 (2025). (Link)

Malyan, V., Kumar, V., Moni, M., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T. S., Fang, J., & Biswas, P., "Assessing the Spatial Transferability of Calibration Models across a Low-cost Sensors Network", *Journal of Aerosol Science*, 181, 106437 (2024). (Link)

Kumar, V., **Malyan, V.**, Sahu, M., & Biswal, B., "Aerosol Sources Characterization and Apportionment from Low-Cost Particle Sensors in an Urban Environment", *Atmospheric Environment: X*, 22, 100271 (2024). (Link)

Malyan, V., Kumar, V., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T. S., Fang, J., & Biswas, P., "Calibrating Low-cost Sensors Using MERRA-2 Reconstructed PM2.5 Mass Concentration as a Proxy", *Atmospheric Pollution Research*, 15(3), 102027 (2023). (Link)

Kumar, V., **Malyan, V.**, Sahu, M., Biswal, B., Pawar, M., & Dev, I., "Spatiotemporal Analysis of Fine Particulate Matter for India (1980–2021) from MERRA-2 Using Ensemble Machine Learning", *Atmospheric Pollution Research*, 14(8), 101834 (2023). (Link)

Kumar, V., **Malyan, V.**, Sahu, M., & Biswal, B., "Machine Learning Classification Model to Label Sources Derived from Factor Analysis Receptor Models for Source Apportionment", *Aerosol and Air Quality Research*, 23(7), 220386 (2023). (Link)

Dharaiya, V. R., **Malyan, V.**, Kumar, V., Sahu, M., Venkataraman, C., Biswas, P., *et al.*, "Evaluating the Performance of Low-cost PM Sensors over Multiple COALESCE Network Sites", *Aerosol and Air Quality Research*, 23(5), 220390 (2023). (Link)

Kumar, A., **Malyan, V.**, & Sahu, M., "Air Pollution Control Technologies for Indoor Particulate Matter Pollution: A Review", *Aerosol Science and Engineering*, 7(2), 261–282 (2023). (Link)

Malyan, V., Kumar, V., & Sahu, M., "Significance of Sources and Size Distribution on Calibration of Low-cost Particle Sensors: Evidence from a Field Sampling Campaign", *Journal of Aerosol Science*, 168, 106114 (2023). (Link)

Kumar, V., **Malyan, V.**, & Sahu, M., "Significance of Meteorological Feature Selection and Seasonal Variation on Performance and Calibration of a Low-Cost Particle Sensor", *Atmosphere*, 13(4), 587 (2022). (Link)

Chapter & Policy Brief

Malyan, V., Kumar, V., Sahu, M., Mayya, Y. S., Biswas, P., & Venkataraman, C., "Improving Air Quality Monitoring in G20 Countries through Low-Cost Sensor–Satellite Synergies", *T20 Policy Brief* (2023).

Sahu, M., **Malyan, V.**, & Mayya, Y. S., "Technologies for Controlling Particulate Matter Emissions from Industries", in *Pollution Control Technologies* (Energy, Environment, and Sustainability), Springer, Singapore (2021). (Link)

Peer Review

Invited **peer reviewer** for **Atmospheric Pollution Research**

Conference Presentations

Oral

"Calibrating Low-cost Sensors Using MERRA-2 Reconstructed PM2.5 Mass Concentration as a Proxy", National Environment Conference (NEC'24), Mumbai, India (2024)

"Optimizing Reference Co-location Sites for Improved Transferability of Calibration Models within a Low-cost Sensor Network in NCT-Delhi", IASTA Conference, Navi Mumbai, India (2023)

"Fundamental Issues with Low-Cost Particle Sensors: Evidence from a Field Sampling Campaign", 2nd Air Sensors International Conference (ASIC), Bangalore, India (2022) – **Student Travel Award**, UC Davis AQRC

"Significance of Meteorological Feature Selection and Seasonal Variation on Performance and Calibration of a Low-Cost Particle Sensor", 12th Asian Aerosol Conference, Virtual (2022)

Poster

"Low-cost Particulate Matter Sensor Evaluation in a Calibration Chamber: Number and Mass Concentration Assessment", 14th Asian Aerosol Conference (AAC'25), Mumbai, India (2025) – lightning talk + poster

"Assessing the Spatial Transferability of Calibration Models across a Low-cost Sensors Network", Air Sensors International Community Conference, Accra, Ghana (2023)

"Distinct Seasonal Effects of Hygroscopicity on the Performance of Low-cost Particle Sensors", 5th India Clean Air Summit (ICAS), Bangalore, India (2023) – **Student Travel Award**, CSTEP

"Significance of Sources and Size Distribution on Calibration of Low-cost Particle Sensors: Evidence from a Field Sampling Campaign", EGU General Assembly, Vienna, Austria (2023)

"Performance Comparison of Different Particulate Matter Control Technologies in Indoor Environments", Indoor Air 2022, Kuopio, Finland (2022)

"Comparative Analysis of Different Indoor Particle Control Technologies", 6th Indian International Conference on Air Quality Management, Virtual (2021)

Teaching Experience

2021 – 2025 | **Teaching Assistant**, Indian Institute of Technology Bombay
Environmental Studies (ES200, Spring 2021), Data Analysis and Interpretation (ES209, 2021/22/24), Air Pollution Control Technologies (ES672, 2022/23/24), Environmental Monitoring Laboratory (ES651, Fall 2023), and Computational Laboratory for Environmental Engineers (ES319, Spring 2025)

PMRF Teaching Assistantship

May 2025 – Jul 2025 | **Guest Lecturer**, Delhi Institute of Rural Development, Delhi, India
Delivered tutorials on Environmental Studies: Science and Engineering to undergrad students

Aug 2024 – Nov 2024 | **Guest Lecturer**, AKS University, Madhya Pradesh, India
Delivered lectures on Environmental Chemistry to postgrad students across a full semester (32 hours), covering the fundamentals of atmospheric chemistry

Jan 2024 – Apr 2024 | **Guest Lecturer**, Sandesh College of Arts, Commerce and Science, Mumbai, India
Delivered lectures on Fundamentals of Organic Chemistry - Reaction Mechanisms to undergrad students across a full semester (36 hours)

Jan 2023 – Apr 2023 | **Guest Lecturer**, Sandesh College of Arts, Commerce and Science, Mumbai, India
Delivered lectures on Calculus II to undergrad students across a full semester (25 hours)

2022 | **Guest Lecture**, Indian Institute of Technology Bombay
Delivered a 1 hour lecture on air pollution control technologies (ESP and baghouse filters) in ES317 Fundamentals of Air Pollution Science and Engineering to BTech + MTech Dual Degree students

Awards & Fellowships

2023 – 25	**Prime Minister's Research Fellowship**, Indian Institute of Technology Bombay, India
2023	**Late Shri Girish Vishnuprasad Desai Award** for Academic Excellence, IIT Bombay, India
2023	**Student Travel Award**, Center for Study of Science, Technology and Policy, India
2022	**ASIC-India Travel Support**, Air Quality Research Center, University of California, Davis, USA
2019 – 22	**Scholarship for MSc–PhD Dual Degree Programme**, Indian Institute of Technology Bombay, India
2016	**Fourth Place**, Intel International Science and Engineering Fair, Phoenix, USA
2015	**First Place**, Initiative for Research and Innovation in STEM (IRIS) National Fair, India
2014	**First Place**, Initiative for Research and Innovation in STEM (IRIS) National Fair, India
2014	**Special Award**, Gujarat Council on Science and Technology, India
2013	**First Place**, Initiative for Research and Innovation in STEM (IRIS) National Fair, India

Personal Projects

Sep 2019 – Apr 2020 | **Installation of Aerators on Taps**, Sustainability Cell, IIT Bombay
- One of five team members on a campus water-conservation initiative to retrofit tap aerators across institute buildings, reducing flow rate without affecting usability
- Conducted **field surveys** to establish baseline consumption and liaised with **institute authorities** to secure approvals and coordinate installation

Aug 2015 – May 2016 | **A Novel Paper Sensor as a Diagnostic Test for Multiple Sclerosis**
- Designed a **low-cost paper-based diagnostic** tool for multiple sclerosis, targeting accurate detection at a fraction of the cost of conventional assays
- Presented at the Intel International Science and Engineering Fair 2016 in Phoenix, USA [Category: Translational Medical Science]

Aug 2014 – May 2016	**GGT Inhibition: Combating Multi-drug Resistance** • Developed a technique to inhibit γ-glutamyl transpeptidase, a candidate pathway in **bacterial multi-drug resistance**, as a route to restoring antibiotic sensitivity • Presented at the Intel International Science and Engineering Fair 2015 in Pittsburgh, USA [Category: Microbiology] • Invited by the Indian Science Congress (ISC'14) and won a special award presented by the Gujarat Council on Science and Technology (GUJCOST)
Aug 2013 – May 2014	**Keratin Waste: An Effective Management** • Converted **keratin waste** from leather industry into a **bio-fertilizer** through microbial degradation by B. licheniformis ER15, diverting a recalcitrant waste stream into agricultural use • Presented at the Intel International Science and Engineering Fair 2014 in Los Angeles, USA [Category: Microbiology]

Skills

Programming	Python, R, MATLAB, LaTeX
Methods	Machine learning, computational fluid dynamics, source apportionment (PMF/NMF)
Software	QGIS, ArcGIS, Origin
Languages	English, Hindi

Academic Service & Positions of Responsibility

May 2023 – Apr 2024	**Department General Secretary**, ESED Council, Indian Institute of Technology Bombay Coordinated the student council and convened technical seminars, cultural events, and industrial visits for the department consisting of 350+ students
Aug 2021 – May 2022	**Alumni Secretary**, ESED Council, Indian Institute of Technology Bombay Coordinated alumni engagement through events and mentorship sessions
Sep 2019 – Jun 2020	**Internship Coordinator**, Institute Placement Cell, Indian Institute of Technology Bombay One of 35 team members serving 1,900+ students; approached companies and universities to source internship opportunities for IIT Bombay students

Organization of Conferences & Professional Memberships

2026 –	**Lifetime Member**, IIT Bombay Alumni Association, India
2024 – 26	**Organizer**, Annual Symposium on Advanced Air Quality Sensing, ANTL, IIT Bombay, India
2024	**Founder and Chief Organizer**, Tatva, ESED Annual Fest, IIT Bombay, India
2023 – 24	**Student Member**, EGU and IASTA

References

PhD Supervisor	**Prof. Manoranjan Sahu** *Associate Professor, Indian Institute of Technology Bombay, India* mrsahu@iitb.ac.in
RPC Member	**Prof. Srinidhi Balasubramanian** *Assistant Professor, Indian Institute of Technology Bombay, India* srinidhi@iitb.ac.in